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An Extension of Bamboo Forest Growth Optimization Algorithm

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***Abstract*—The Bamboo Forest Growth Optimization Algorithm (BFGO) is an innovative and flexible meta-heuristic optimization algorithm inspired by the growth pattern of bamboo forests, which first establish roots and extend before growing freely. This algorithm has achieved good results in solving optimization and engineering problems. However, it suffers from insufficient global exploration capabilities and is prone to getting stuck in local optima. Drawing on additional growth characteristics of bamboo forests, this paper extends BFGO's search strategy, reconstructs the bamboo growth model, and designs an elimination strategy to enhance population quality. Comparative experiments are conducted on the CEC 2017 and CEC 2013 benchmark suites against the original algorithm, state-of-the-art algorithms, classical algorithms and improved variants, followed by ablation studies. The results demonstrate that the extended BFGO algorithm achieves superior performance, effectively validating the efficacy and competitiveness of the proposed enhancements.**

***Index Terms*—Bamboo forest growth optimization algorithm, Swarm intelligence algorithm, Logistic growth model, Natural selection strategy**

I. Introduction

In recent years, research and applications related to meta-heuristic algorithms [1]–[5] have developed rapidly [6] and are widely used in various optimization problems [7], [8]. These algorithms are inspired by some real-life phenomena or laws so as to obtain an approximate optimal solution to the problem by iterative search within a given problem horizon. Compared with the traditional mathematical methods which are limited by strong constraints such as the differentiability and continuity of the objective function, these algorithms can continuously search in non-convex, high-dimensional, or discrete-continuous solution spaces by defining the fitness function, and have the advantages of weak dependence on the problem domain, high adaptability, scalability and robustness to complex constraints, and thus they can deal with a wide range of large-scale and complex problems. On the other hand, bamboo forest growth optimization algorithm(BFGO) is inspired by the physiological properties and growth process of bamboo [9]. Bamboos, as herbaceous plants, can grow to the height of trees, which is highly related to their unique growth pattern [10], [11]. The "slow-fast-slow" growth cycle of bamboos and their rapid stem elongation contribute to their ecological dominance: bamboos first use their root system to grow and spread in random directions over several kilometers above the ground in order to continuously absorb nutrients from the soil for optimal growth. Bamboo shoots emerging from the soil can grow very long in a short period of time. Therefore, Feng et al. mathematically modeled the two phases of complete extension of bamboo whips and rapid growth of bamboo and mapped these two growth processes in an optimization algorithm. In the bamboo whip extension stage, individuals on each population move in different directions to facilitate global exploration of the solution space; in the bamboo growth stage, each individual is locally exploited according to the constructed growth model so as to update the position. This is the overall framework of the bamboo forest growth optimization algorithm.

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In Feng's work [9], BFGO algorithm is different from the classical meta-heuristic algorithm, it is inspired by the special growth law of bamboo, and carries out global exploration and local search in stages, which is innovative and performs well in the practice of optimization problems with other algorithms for engineering applications, which shows that it is quite competitive in dealing with complex problems. However, its performance on the CEC 2017 test set [12] is not outstanding, and its algorithmic mechanisms reveal room for improvement in both exploration and exploitation capabilities.

In the BFGO algorithm, the study of bamboo growth characteristics remains insufficiently comprehensive, leaving considerable room for incorporating more natural design. This paper proposes an extended bamboo forest growth optimization algorithm (E-BFGO), which aims to address the limitations of BFGO. In the natural growth process of bamboo forests, in addition to the unique growth characteristics of bamboo forests themselves and the iterative law, it will be accompanied by light, water, soil nutrients and other external environmental factors that affect growth. Bamboo has a well-developed root system, which is concentrated underground to form a dense root network in the early stage. If the soil is in deep and loose fertile sandy loam or light loam, it can better satisfy the demand of its root system growth. After the bamboo whip absorbs nutrients in the soil for months or even years to form shoots, the bamboo shoots rely on light and store nutrients to grow rapidly, produce branches and leaves, and wait until the new shoots of the bamboo whip regenerate to complete the population iteration. This paper draws inspiration from this and simulates ecological features of bamboo to supplement and refine the original algorithm with additional strategies, while replacing and optimizing its flawed growth mechanisms. More regulatory factors from bamboo growth processes are integrated and analyzed in conjunction with algorithmic mechanisms. By employing these extended strategies, the BFGO algorithm's structure is constructed and enhanced. The new strategies incorporate both broad exploration and local exploitation methods during the bamboo whip extension and bamboo shoot growth stages, followed by a natural population replacement mechanism. These approaches collectively assist the algorithm in escaping local optima, improving both global and local search capabilities, and enhancing overall solution quality.

II. BFGO

This section will detail the Bamboo Forest optimization algorithm, the inspiration for the expansion strategy, and their mathematical models.

In the initial stage of the algorithm, the population is partitioned into K clusters, each consisting of n individuals. Following a simplified initialization, the search procedure commences. The core of the BFGO algorithm lies in referencing the common growth habits of bamboo plants, including the exploration of underground stem extension and rapid growth of above ground parts, and then analyzing its mathematical model to transform it into a search strategy in the algorithm, with two distinct phases: broad exploration and local search.

During the bamboo whip extension stage, the individuals of each cluster move in the main direction with three benchmarks for broad exploration: the group cognitive term, the group memory term and the group center term. The specific movement equations are shown as:

$$X_{t+1} = \begin{cases} X_G + Q \times (\sigma \times X_G - X_t) \times cos\alpha, & r_1 < 0.4 \\ X_P(k) + Q \times (\sigma \times X_P(k) - X_t) \times cos\beta, & 0.4 \le r_1 < 0.7 \\ C(k) + Q \times (\sigma \times C(k) - X_t) \times cos\gamma, & else \end{cases} \tag{1}$$

$$cos\alpha = \frac{X_t \cdot X_G}{|X_t| \times |X_G|} \tag{2}$$

$$cos\beta = \frac{X_t \cdot X_P(k)}{|X_t| \times |X_G|} \tag{3}$$

$$cos\gamma = \frac{X_t \cdot C(k)}{|X_t| \times |X_G|} \tag{4}$$

$$Q = 2 - \frac{t}{T} \tag{5}$$

Where X_G represents the individual with the best fitness among all individuals, $X_P(k)$ represents the individual with the best fitness within the k-th bamboo whip cluster, and $C(k)$ represents the central position of the k-th bamboo whip cluster, typically calculated as the average of the positions of all individuals within the cluster. α, β, and γ denote the three extension directions of the rhizome system, namely, the direction of the current individual moving towards the global optimal solution, the population optimal solution, and the center of the population. In addition, σ is a random number between 1 and 2. In Eq. (5), Q denotes the moving step size that decreases from 2 to 1 as the number of iterations t increases.

BFGO simulates the behavior of bamboo whips randomly lengthening and expanding their territories underground on this stage, and guides the population to explore extensively in the solution space through a multi-directional and large-scale search mechanism to avoid the algorithm from falling into local optimal solutions. However, Feng et al. adopted a conservative approach in designing the moving step size, bamboo whip individuals lack the long-range jumping ability, and the parameter σ is inappropriately applied to enhance directional targeting of individuals. Overall, the search mechanism exhibits limited flexibility.

During the bamboo shoot growth stage, a temporary population is defined to replace the original population, which is defined by the following equations:

$$X_D = 1 - \left| \frac{X_t - C(k) + 1}{X_G - C(k) + 1} \right| \tag{6}$$

$$q(t) = X_G \times e^{-d} \times e^{\frac{b}{\varphi \times t^{\varphi}}} \tag{7}$$

$$\Delta H = \frac{q(t) - q(t-1)}{X_G - X_t} \tag{8}$$

$$X_{temp} = \begin{cases} X_t + X_D \times \Delta H \\ X_t - X_D \times \Delta H \end{cases} \tag{9}$$

Where X_D denotes the relationship between the distance of the particle to the population center position and the distance of the best individual of the population to the population center position, $q(t)$ denotes the cumulative growth of the t-th iteration, and ΔH denotes the incremental increase of one iteration. d is a parameter in the range of (-1, 1), and both b and φ are fixed parameters representing the site conditions of the bamboo.

Eq. (7) is derived using the differential equation for the variation of bamboo height with growth [13] derived by Sloboda [14] and the integral form and its simplified equation to give, which can be expressed as follow:

$$\frac{dy}{dt} = \frac{m \times y}{t^l ln\left(\frac{n}{y}\right)} \tag{10}$$

$$y = n \times e^{-C} \times e^{\frac{m}{(l-1)\times t^{(l-1)}}} \tag{11}$$

$$y = SI \times e^{\frac{m}{k \times t^k}} \tag{12}$$

Where l, m, n are the parameters of the growth model, and ensure that l >1, m >0, n >0, t represents the growth time, y is the growth height of bamboo shoots. C is an integral constants. SI represents the maximum achievable bamboo height, which varies with changes in site conditions, and k is the alternative parameter after model simplification.

This stage of the BFGO algorithm simulates the process of bamboo shoots breaking through the soil and growing at a high rate, utilizing a height growth model to update individual positions in order to enhance explorability and undertake the task of local exploitation of the current solution. However, Eq. (8) indicates that the individual update step size depends on the gain difference relative to the optimal value. If the algorithm becomes trapped in a flat region, individual updates may stagnate. Furthermore, the movement of individuals in this stage lacks both directionality and randomness, leading to chaotic searches around the current position. Overall, the adaptability of the growth model is limited, and the algorithm's local exploitation capability requires improvement.

III. E-BFGO

A. Initialization

In this stage, the population initialization method, which is not thoroughly addressed in BFGO, will be optimized to ensure high-quality initial populations for subsequent global search. In BFGO algorithm, the bamboo whip and bamboo shoot populations are integrated, whereas in this extended version, they should be decoupled and separated to conduct global and local searches independently, thereby enhancing the overall optimization capability. Additionally, each individual is assigned an age attribute L_t that serves as a regulatory factor in subsequent stages, with its value incremented by 1 at the end of each iteration. Finally, each of the two types of individuals will be divided into K clusters, and the initial values of the individuals are mapped into the solution space by generating chaotic sequences successively from the Logistic mapping equations [15], [16], the form of the equations as well as the population initialization are designed as shown in Eqs. (13)–(14):

$$w_{n+1} = \ cr \times w_n \times (1 - w_n) \tag{13}$$

$$x_n = \ x_{min} + \ w_n \times (x_{max} - x_{min}) \tag{14}$$

Where cr is a control parameter, w_n denotes the n-th chaotic sequence value, and x_n is the initial value of the n-th individual. This initialization method the population can be made more uniformly distributed in the solution space with higher randomness, ensuring diversity of the initial population across the search space and increasing the likelihood of discovering high-quality solutions.

B. Extended Bamboo Whip Extension Stage

This stage focuses on improving and extending the shortcomings of the original BFGO algorithm in bamboo whip extension stage.

First, the original extension exploration formula needs to be reformulated. A new movement step size S will be introduced to cooperatively control the exploration distance of the bamboo whip individuals, thereby expanding the breadth of search while allowing dynamic adjustment of the search amplitude to escape local basins of attraction. Moreover, as the iteration progresses, the search amplitude should be appropriately reduced and focused on the region near the target for precise search. In nature, the bamboo whip, as the underground stem system of bamboo, is densely covered with absorbing roots that extract water and mineral nutrients from the soil to support its rapid extension. Similarly, the age of the bamboo can also affect its extension behavior. Before a bamboo shoot emerges above ground, its root system actively extends to acquire more nutrients; once the shoot breaks through the soil, the whip activity gradually ages. Inspired by this, the soil nutrient index and individual age can be defined as influential variables in the search process, promoting extensive exploration and enabling flexible adjustment of step sizes. Based on these concepts, the formula for S is designed as:

$$S = (0.7 \times SN + 0.3 \times \frac{T_s}{L_t}\ )^Q \tag{15}$$

$$SN = BN \times e^{-0.5 \frac{t}{T}} \tag{16}$$

Where SN represents the soil nutrient index, while BN denotes the soil base nutrient. SN decays over time relative to the baseline BN to regulate the search span. T_s and L_t denote the standard time for bamboo shoot emergence and the current age of the individual, respectively. These two factors can generate substantial momentum for individual exploration when the individuals are young or during the early stages of the algorithm. As individuals mature or in later iterations, they progressively reduce the convergence step size, enabling a smooth transition from explosive exploration to slowly exploitation. Subsequently, the exploration formula for the bamboo whip individuals is reformulated as:

$$X_{t+1} = \begin{cases} X_G + Q \times S \times (X_G - X_t) \times cos\alpha, & r_1 < 0.4 \\ X_P(k) + Q \times S \times (X_P(k) - X_t) \times cos\beta, & 0.4 \le r_2 < 0.7 \\ C(k) + Q \times S \times (C(k) - X_t) \times cos\gamma, & else \end{cases} \tag{17}$$

In Eq. (17), the original search step size is replaced, and the less influential parameter σ is removed to enhance the exploration performance of the bamboo whip individuals. Additionally, the movement mechanism based on three primary references tends to trap the algorithm in local optima. To improve the global convergence accuracy during the extension phase, random perturbations are applied after the position updates of the bamboo rhizome individuals, thereby strengthening their local search capabilities, enabling them to identify individuals with better fitness within the neighborhood. The specific form of the perturbation is shown in Eq. (18):

$$X_{Wt}' = X_{Wt} + \delta, \delta \in \left[\frac{X_{min}}{5 \times (X_{max} - X_{min})}, \frac{X_{max}}{5 \times (X_{max} - X_{min})}\right] \tag{18}$$

C. Extended Bamboo Shoot Growth Stage

In this stage, the original bamboo shoot growth strategy will be replaced with a more effective growth model for exploration and local exploitation.

Shi et al. [13] used the growth data of 90 typical Moso bamboo samples to give the measured growth curve of bamboo shoots, as shown in Fig. 1.

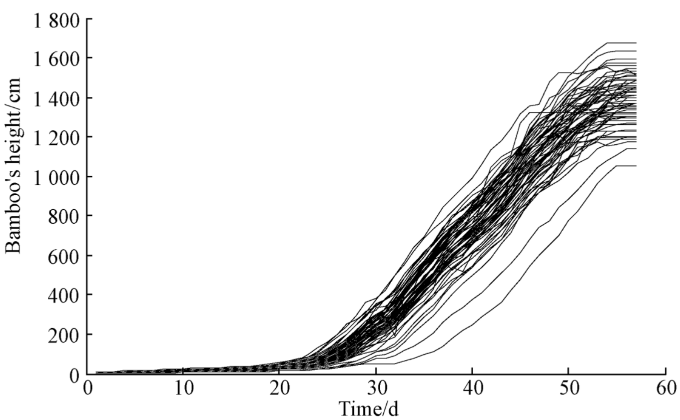

Fig. 1. Bamboo Shoots Growth Curve

We can find from the figure that the growth of this Moso bamboo shoot span a total of 57 days, and can be divided into three distinct stages: an initial phase of slow growth, followed by a stage of explosive growth, and finally a third stage where growth stabilizes. If modeled using Eq. (12), the growth pattern of the third stage cannot be accurately captured; instead, the model predicts continued growth. Logistic growth model [16] is one of the most widely used models to describe the changes in biomass and body length data of plants and animals, and it is highly compatible with the "S-shaped growth curve" of the individual height of bamboo shoot, which shows a "slow-fast-slow" trend [17]. Fig. 2 compares our Logistic growth model and the original model, both constructed through fitting based on data from Fig. 1. By extending the growth period to 80 days, a clear differentiation between them is demonstrated.

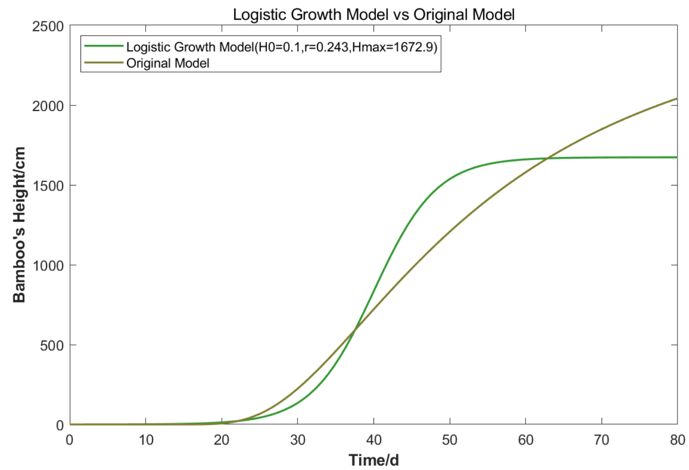

Fig. 2. Logistic Growth Model vs Original Model

Moreover, the adoption of the Logistic growth model enables adaptive individual movement toward the target, establishing a positive correlation between the movement span and the distance to the target. It achieves a balance between exploring potential high-quality solutions and performing local exploitation within the target region, thereby endowing the algorithm with greater flexibility in its search process. Clearly, the Logistic growth model is more appropriate for capturing the growth traits of bamboo. subsequently, Sloboda's bamboo height growth model is replaced in this paper with a logistic model, whose differential form and equation for bamboo shoot individual growth are as follows:

$$\frac{dh}{dt} = r \times h \times \frac{H_{max} - h}{H_{max}} \tag{19}$$

$$X_{St+1} = X_{St} + (0.5 \times LN + 0.5 \times SN) \times \left(\frac{X_G - X_{St}}{\|X_G - X_{St}\|} \odot |X_{St} - C(k)|\right) \times \frac{\|X_G - X_{St}\|}{\|X_G - C(k)\|} \tag{20}$$

In Eq. (19), r represents the growth rate, h denotes the current height of the individual bamboo plant, and H_{max} is the maximum achievable height. $r \times h$ indicates the individual's current growth capacity, while $\frac{H_{max} - h}{H_{max}}$ reflects the degree of growth resistance encountered.

In Eq. (20), the global optimum position X_G guides individual movement, then $X_G - C(k)$ represents H_{max} and

$X_S - C(k)$ represents h. The growth model in Eq. (19) is designated as the baseline increment for individual movement. The Light Index LN and Soil Nutrient Index SN collectively serve as the motion increment for this stage, simultaneously fulfilling the role of r. This design is informed by the core regulatory functions of light and nutrients in plant growth. Sufficient light and nutrients not only promote the accumulation of organic substances such as carbohydrates but also provide the energetic foundation for bamboo shoot germination and growth. The calculation of LN is as Eq. (21):

$$LN(k) = GL \times \frac{rank(k)}{K} \tag{21}$$

30 78 Where GL represents the global light index, which is taken randomly at each iteration. $rank(k)$ is the mean fitness ranking of the k-th cluster where an individual is located among the K clusters. This allows cluster with poorer fitness to receive more light, thereby increasing the movement energy of their individuals and preventing them from being trapped in inferior regions for extended periods. Simultaneously, high-quality cluster located closer to the optimal solution can focus on exploiting their nearby areas.

40 37 83 In addition, to improve search precision and randomness, local exploitation of the neighborhood around the current high-quality solution is required. Wang et al. modeled bamboo according to its natural morphological structure using fractal graphic techniques [18], and they found that bamboo has typical fractal features, the ratio of the angle between most bamboo branches and the growth axis to the right angle approximates the golden section ratio, and they constructed the rotation matrix of the branches in three dimensions. A fractal model for individual bamboo shoots is accordingly designed to simulate the branching during the growth of bamboo joints in Eqs. (22)–(25):

5

$$X_{S_t}' = \begin{cases} X_{S_t} + randn \times R \times \begin{bmatrix} bl \\ 0 \end{bmatrix}, & r_2 < 0.3 \\ X_{S_t} + randn \times R \times \begin{bmatrix} 0 \\ bl \end{bmatrix}, & 0.3 \le r_2 < 0.6 \\ X_{S_t}, & else \end{cases} \tag{22}$$

26

$$R = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \tag{23}$$

$$\theta = 34.14 \times \frac{\pi}{180} \tag{24}$$

$$bl = (1 - \frac{L_t + 1 - T_s}{T_d + 1 - T_s}) \times \left\| \frac{\Delta h_t}{max(\Delta h_t)} \right\| \tag{25}$$

95 75 Where R is the rotation matrix, θ is the rotation angle, Δh represents the height of the bamboo joint, i.e. the increment of movement during growth. After normalization, it interacts with the bamboo shoot age factor to form the modulus of the branch bl, where T_d denotes the dead time. Young bamboo shoots typically grow longer joint, implying that the individual's exploitation distance decreases with age. As individuals age, they gradually approach high-quality solutions and adjust their local search space accordingly. The growing bamboo shoot individuals will have a higher probability of growing branches in multiple random two-dimensional directions, and the excellent individuals with bamboo branches will replace the normal ones, while there will also exist some bamboo shoot individuals without bamboo branches.The branching strategy establishes a stochastic and dynamically adjusted multi-directional exploration field, thereby strengthening the algorithm's local exploitation capability.

D. *Natural Selection Mechanism*

As the number of iterations increases, the algorithm is prone to homogenization, leading to a loss of population diversity. At this point, continued iteration fails to yield effective improvements in the solution. Therefore, a population maintenance strategy is considered. The Forest Optimization Algorithm (FOA) developed a selection mechanism for population renewal and elite retention [19], which ranked candidate solutions from best to worst in terms of fitness, removed trees exceeding the age limit and selects superior solutions. By referring to this, at the end of each iteration, natural selection is performed on the whole population. The individuals in each cluster are sorted by fitness value, and a certain number of weak individuals are proportionally eliminated within the cluster based on a set elimination rate α. In addition, individuals whose age reaches a set number of years are eliminated.If the eliminated individuals have already grown bamboo shoots, they will be replaced by new individuals, and the generation of new individuals will follow the initialization method. In this way, a natural replacement is accomplished, retaining the young elite individuals and avoiding a certain amount of poor quality solutions that continue to occupy resources, thereby aiding the algorithm in escaping local optimum regions. It can also synergize with the extended extension strategy, enabling new individuals to maintain broad search even during the middle and later stages of the algorithm, thereby facilitating the discovery of potentially high-quality solutions. This makes it an essential component within the overall framework. The clearer steps can be viewed from the pseudo-code.

E. *Algorithmic Procedure*

The pseudo-code of the E-BFGO algorithm as shown in Algorithm 1.

Algorithm 1: Pseudocode of E-BFGO

Input: N:population size, D:dimension, It:the number of function iterations, $[lb, ub]$:problem boundary

Output: the best solution X_G

Initialize number of clusters K, base soil nutrition BN, age of bamboo L_n, time of growing into bamboo shoot T_s, time of death T_{end}, elimination rate α.
Initialize the bamboo whip position X_W_t and the bamboo shoot position X_S_t using Eqs. (13)–(14), then divide the bamboo whip population into K clusters in alternating order based on fitness and replicate them to the bamboo shoot population
while t <*It* **do**
  if X_G *not updated* **then**
    select some elite individuals and poor individuals to update X_W_t, X_S_t, and update X_P, X_G.
  end if
  if X_P *not updated* **then**
    redivide the population in alternating order.
  end if
  Set soil nutrition SN based on Eq. (15).
  Update X_W_t according to Eqs. (16)–(17), and add disturbance to X_W_t based on Eq. (18).
  Update X_P and X_G.
  if $L_n \geq T_s$ **then**
    Sort bamboo shoot clusters according to the average fitness, set global light GL.
    Update X_S_t according to Eqs. (20)–(21).
    while $d \leq D$ **do**
      using Eqs. (22)–(25) to make X_S_t grow branches.
      d = d + 2.
    end while
    if *r3 <0.8* **then**
      Select X_S_t with branches.
    else
      Select X_S_t without branches.
    end if
  end if
  Sort individuals within cluster accoring to their fitness, and elimitate $\alpha \times N$ poor individuals.
  Eliminate individuals with L_n greater than or equal to T_{end}.
  Generate new individuals using Eqs. (13)–(14) to replace eliminated individuals.
  Update X_P and X_G.
  t = t + 1, L_n = L_n + 1, and redivide the population into K clusters in alternating order based on fitness.
end while

IV. Experiment and Analysis

This section will conduct a comprehensive evaluation of the optimization performance of the E-BFGO algorithm, firstly comparing and analyzing it against the original algorithm, several state-of-the-art algorithms, classical algorithms, and their improved variants, including PSO [20], VPPSO [21], SCA [22], EDOLSCA [23], BKA [24], PLO [25], DE [26]. Table I lists the parameter settings for each algorithm.

TABLE I
Algorithms and Parameter Settings

Algorithms	Parameters
PSO	$c_1 = 2, c_2 = 2, w_{max} = 0.9$
VPPSO	$c_1 = 1.5, c_2 = 1.5$
SCA	$a = 2$
EDOLSCA	$a = 2, Jr = 0.5, w = 12$
BKA	$p = 0.9$
PLO	$m = 100$
DE	$F = 0.4, Cr = 0.9$
BFGO	$n = 5, \varphi = 2$
E-BFGO	$cr = 4, n = 5, \alpha = 0.2, BN = 2, T_s = 3, T_d = 8$

In the comparative experiment, a total of 30 benchmark functions from CEC2017 and 28 benchmark functions from CEC2013 [27] are employed for testing. These two test sets encompass function types of varying difficulty and characteristics: unimodal functions (F1–F3 in CEC 2017; F1–F5 in CEC 2013), basic multimodal functions (F4–F10 in CEC 2017; F6–F20 in CEC 2013), hybrid functions (F11–F20 in CEC 2017), and composition functions (F21–F30 in CEC 2017; F21–F28 in CEC 2013), which can comprehensively calibrate the algorithm's ability to deal with various problems.

The search range of all test functions is [-100,100], the search dimensions is set to 30 dimensions, the population size is set to 30, and each algorithm is independently experimented 30 times. To ensure the test results are fair, the maximum number of evaluations for all algorithms is set to 15000.

Table II show the mean, standard deviation, and median values of each algorithm on the CEC2017 test set, and table III show the corresponding indexes on the CEC2013 test set, respectively. Where the optimal results are bold, and the mean values are accompanied by significance analysis results obtained using the Wilcoxon rank-sum test ("+" indicates that E-BFGO is significantly superior, "=" indicates no significant difference, "-" indicates that E-BFGO is significantly inferior).

Subsequently, to assess the effectiveness of the extended strategies in the E-BFGO algorithm, variant versions are constructed by separately removing the extended extension strategy (EBFGO-DeEBWES), the extended growth strategy (EBFGO-DeEBSGS), and the natural selection mechanism (EBFGO-DeNSM). These variants are then compared experimentally with the original BFGO algorithm. All algorithms are tested under the same conditions as in the previous experiments, with independent runs conducted accordingly. Fig. 3 illustrates their convergence processes on the CEC 2017 test set, along with the logarithmic form of the difference between their results and the standard function values. For convenience, two test results are selected from each of the four function types as representative samples.

TABLE II
COMPARISON OF TEST RESULTS OF VARIOUS ALGORITHMS ON 30 DIMENSIONS OF CEC2017

Test function	Index	PSO	VPPSO	SCA	EDOLSCA	BKA	PLO	DE	BFGO	E-BFGO
F1	Mean	1.42E+10(+)	1.12E+08(=)	2.14E+10(+)	1.01E+10(+)	1.47E+10(+)	1.81E+09(+)	4.57E+08(+)	5.81E+09(+)	**6.10E+07**
	Std	2.14E+09	2.04E+08	4.14E+09	5.14E+09	1.14E+10	3.72E+08	6.31E+08	3.40E+09	**4.43E+07**
	Med	1.43E+10	5.56E+07	2.18E+10	9.96E+09	1.29E+10	1.78E+09	2.42E+08	4.91E+09	**4.82E+07**
F2	Mean	2.56E+34(+)	4.63E+27(=)	2.75E+38(+)	2.92E+35(+)	1.25E+42(+)	1.66E+28(+)	6.05E+29(+)	1.78E+38(+)	**2.80E+24**
	Std	7.62E+34	2.48E+28	7.64E+38	1.28E+36	6.80E+42	3.28E+28	3.16E+30	9.72E+38	**1.03E+25**
	Med	4.84E+33	2.13E+23	3.82E+36	2.89E+31	4.47E+29	3.07E+27	2.54E+25	1.45E+32	**3.12E+22**
F3	Mean	1.34E+05(+)	6.12E+04(+)	9.52E+04(+)	1.18E+05(+)	4.61E+04(+)	1.56E+05(+)	1.02E+05(+)	6.59E+04(+)	**3.29E+04**
	Std	3.73E+04	1.27E+04	1.79E+04	2.92E+04	1.75E+04	2.54E+04	2.60E+04	9.15E+03	**8.70E+03**
	Med	1.25E+05	6.18E+04	9.34E+04	1.20E+05	4.09E+04	1.58E+05	9.74E+04	6.73E+04	**3.19E+04**
F4	Mean	1.46E+03(+)	5.63E+02(=)	3.07E+03(+)	1.78E+03(+)	2.40E+03(+)	6.14E+02(+)	6.12E+02(+)	1.38E+03(+)	**5.61E+02**
	Std	1.84E+02	3.60E+01	8.78E+02	8.41E+02	3.19E+03	**3.30E+01**	9.19E+01	4.69E+02	3.85E+01
	Med	1.42E+03	5.63E+02	2.93E+03	1.53E+03	1.21E+03	6.12E+02	5.85E+02	1.28E+03	**5.52E+02**
F5	Mean	7.81E+02(+)	6.79E+02(+)	8.29E+02(+)	7.75E+02(+)	7.77E+02(+)	6.83E+02(+)	7.22E+02(+)	7.66E+02(+)	**6.39E+02**
	Std	**1.34E+01**	4.50E+01	2.48E+01	3.19E+01	5.82E+01	1.45E+01	2.36E+01	5.69E+01	3.72E+01
	Med	7.81E+02	6.72E+02	8.31E+02	7.70E+02	7.67E+02	6.83E+02	7.22E+02	7.61E+02	**6.40E+02**
F6	Mean	6.42E+02(+)	6.45E+02(+)	6.64E+02(+)	6.53E+02(+)	6.64E+02(+)	6.30E+02(+)	**6.04E+02(-)**	6.58E+02(+)	6.25E+02
	Std	4.98E+00	8.26E+00	6.44E+00	1.19E+01	9.69E+00	3.52E+00	**2.71E+00**	7.33E+00	8.07E+00
	Med	6.41E+02	6.46E+02	6.62E+02	6.51E+02	6.63E+02	6.31E+02	**6.03E+02**	6.58E+02	6.26E+02
F7	Mean	1.52E+03(+)	1.05E+03(+)	1.24E+03(+)	1.22E+03(+)	1.28E+03(+)	1.09E+03(+)	9.81E+02(+)	1.11E+03(+)	**9.14E+02**
	Std	8.52E+01	9.14E+01	5.74E+01	8.10E+01	7.77E+01	3.61E+01	**2.55E+01**	7.76E+01	4.33E+01
	Med	1.52E+03	1.05E+03	1.22E+03	1.22E+03	1.29E+03	1.10E+03	9.77E+02	1.11E+03	**9.01E+02**
F8	Mean	1.09E+03(+)	9.44E+02(+)	1.10E+03(+)	1.06E+03(+)	1.01E+03(+)	9.75E+02(+)	1.01E+03(+)	1.02E+03(+)	**9.22E+02**
	Std	**1.46E+01**	2.56E+01	2.28E+01	3.19E+01	4.64E+01	1.93E+01	2.36E+01	3.73E+01	2.28E+01
	Med	1.08E+03	9.42E+02	1.11E+03	1.06E+03	1.00E+03	9.75E+02	1.01E+03	1.02E+03	**9.26E+02**
F9	Mean	6.87E+03(+)	4.68E+03(+)	8.48E+03(+)	9.42E+03(+)	5.80E+03(+)	4.95E+03(+)	**1.59E+03(-)**	6.23E+03(+)	2.86E+03
	Std	9.35E+02	9.06E+02	2.23E+03	3.47E+03	1.10E+03	1.24E+03	**6.32E+02**	1.65E+03	9.36E+02
	Med	6.93E+03	4.72E+03	8.38E+03	8.88E+03	5.91E+03	4.73E+03	**1.37E+03**	6.18E+03	2.69E+03
F10	Mean	8.76E+03(+)	5.08E+03(+)	8.82E+03(+)	7.89E+03(+)	6.28E+03(+)	6.14E+03(+)	9.14E+03(+)	7.11E+03(+)	**4.65E+03**
	Std	4.49E+02	6.94E+02	4.18E+02	6.79E+02	1.61E+03	4.15E+02	**3.50E+02**	7.37E+02	6.67E+02
	Med	8.73E+03	5.06E+03	8.87E+03	7.96E+03	5.71E+03	6.25E+03	9.25E+03	7.02E+03	**4.57E+03**
F11	Mean	3.97E+03(+)	1.63E+03(+)	3.91E+03(+)	3.35E+03(+)	2.16E+03(+)	2.58E+03(+)	1.57E+03(+)	2.47E+03(+)	**1.30E+03**
	Std	7.12E+02	2.83E+02	1.14E+03	1.46E+03	1.24E+03	5.37E+02	6.20E+02	7.95E+02	**5.01E+01**
	Med	3.89E+03	1.60E+03	3.65E+03	2.72E+03	1.75E+03	2.53E+03	1.38E+03	2.26E+03	**1.29E+03**
F12	Mean	1.23E+09(+)	3.82E+07(+)	2.44E+09(+)	1.15E+09(+)	7.06E+08(+)	2.57E+07(+)	**7.43E+06(-)**	3.37E+08(+)	1.17E+07
	Std	2.91E+08	3.41E+07	7.08E+08	1.49E+09	1.66E+09	**9.00E+06**	9.31E+06	4.31E+08	9.98E+06
	Med	1.17E+09	2.97E+07	2.29E+09	5.42E+08	7.74E+07	2.41E+07	**5.93E+06**	1.97E+08	9.71E+06
F13	Mean	4.55E+08(+)	1.08E+05(+)	1.26E+09(+)	5.93E+08(+)	9.47E+07(+)	2.60E+05(+)	7.01E+04(=)	4.64E+07(+)	**5.68E+04**
	Std	1.47E+08	7.08E+04	4.71E+08	1.63E+09	4.45E+08	9.88E+04	7.16E+04	7.55E+07	**6.95E+04**
	Med	4.53E+08	9.23E+04	1.12E+09	2.23E+07	2.23E+05	2.49E+05	4.86E+04	1.76E+07	**4.20E+04**
F14	Mean	6.21E+05(+)	2.58E+05(+)	8.65E+05(+)	7.51E+05(+)	1.48E+05(=)	7.97E+04(+)	5.40E+04(+)	6.73E+05(+)	**3.49E+04**
	Std	3.92E+05	2.30E+05	5.33E+05	7.08E+05	6.23E+05	6.32E+04	1.22E+05	6.79E+05	**3.90E+04**
	Med	5.43E+05	2.49E+05	6.89E+05	4.94E+05	1.59E+04	4.79E+04	**6.52E+03**	4.71E+05	2.15E+04
F15	Mean	8.49E+07(+)	4.51E+04(+)	5.29E+07(+)	1.00E+07(+)	9.80E+06(+)	2.44E+04(+)	1.50E+04(=)	1.24E+06(+)	**1.08E+04**
	Std	4.68E+07	3.26E+04	5.11E+07	4.29E+07	5.34E+07	9.64E+03	1.73E+04	3.35E+06	**9.30E+03**
	Med	8.46E+07	3.76E+04	4.15E+07	9.62E+05	3.92E+04	2.38E+04	8.12E+03	2.54E+05	**7.03E+03**
F16	Mean	3.82E+03(+)	2.90E+03(=)	4.09E+03(+)	3.51E+03(+)	3.35E+03(+)	**2.72E+03(=)**	3.59E+03(+)	3.51E+03(+)	2.82E+03
	Std	2.27E+02	2.82E+02	3.39E+02	3.44E+02	4.97E+02	2.20E+02	**2.17E+02**	5.39E+02	2.69E+02
	Med	3.79E+03	2.89E+03	4.13E+03	3.55E+03	3.25E+03	**2.71E+03**	3.58E+03	3.41E+03	2.77E+03
F17	Mean	2.72E+03(+)	2.26E+03(=)	2.85E+03(+)	2.43E+03(=)	2.54E+03(+)	**2.03E+03(-)**	2.49E+03(+)	2.59E+03(+)	2.29E+03
	Std	1.44E+02	1.61E+02	1.75E+02	3.12E+02	3.06E+02	**1.04E+02**	2.04E+02	3.36E+02	2.08E+02
	Med	2.75E+03	2.28E+03	2.89E+03	2.40E+03	2.48E+03	**2.02E+03**	2.52E+03	2.58E+03	2.28E+03
F18	Mean	8.67E+06(+)	1.49E+06(+)	1.62E+07(+)	6.13E+06(+)	3.98E+05(=)	8.20E+05(+)	**1.88E+05(-)**	3.77E+06(+)	4.39E+05
	Std	4.75E+06	1.32E+06	1.01E+07	8.36E+06	4.78E+05	5.13E+05	**1.91E+05**	4.76E+06	4.10E+05
	Med	8.66E+06	1.17E+06	1.43E+07	4.03E+06	2.75E+05	6.39E+05	**1.19E+05**	2.28E+06	3.36E+05
F19	Mean	1.02E+08(+)	2.14E+06(+)	1.18E+08(+)	2.42E+07(+)	3.22E+06(+)	5.40E+04(+)	**9.71E+03(=)**	4.30E+06(+)	1.45E+04
	Std	3.53E+07	1.09E+06	1.11E+08	5.78E+07	8.97E+06	4.83E+04	**7.85E+03**	3.51E+06	1.34E+04
	Med	1.07E+08	2.06E+06	8.09E+07	5.66E+06	3.49E+05	4.37E+04	**7.22E+03**	3.50E+06	8.71E+03
F20	Mean	2.87E+03(+)	2.64E+03(+)	2.94E+03(+)	2.78E+03(+)	2.70E+03(+)	**2.45E+03(=)**	2.88E+03(+)	2.79E+03(+)	2.50E+03
	Std	1.54E+02	2.00E+02	1.89E+02	1.91E+02	1.55E+02	**9.94E+01**	2.61E+02	1.64E+02	1.90E+02
	Med	2.88E+03	2.62E+03	2.98E+03	2.81E+03	2.73E+03	**2.45E+03**	2.90E+03	2.80E+03	2.49E+03
F21	Mean	2.56E+03(+)	2.46E+03(+)	2.61E+03(+)	2.54E+03(+)	2.58E+03(+)	2.46E+03(+)	2.51E+03(+)	2.58E+03(+)	**2.42E+03**
	Std	1.83E+01	3.16E+01	2.96E+01	3.08E+01	7.25E+01	**1.66E+01**	1.71E+01	6.39E+01	3.00E+01
	Med	2.56E+03	2.46E+03	2.61E+03	2.54E+03	2.57E+03	2.46E+03	2.51E+03	2.56E+03	**2.42E+03**
F22	Mean	8.06E+03(+)	5.36E+03(+)	9.95E+03(+)	6.19E+03(+)	6.63E+03(+)	3.54E+03(+)	8.58E+03(+)	7.67E+03(+)	**3.53E+03**
	Std	2.97E+03	2.06E+03	1.40E+03	2.64E+03	1.45E+03	**1.33E+03**	3.22E+03	1.83E+03	1.87E+03
	Med	9.93E+03	6.19E+03	1.02E+04	4.73E+03	6.82E+03	3.07E+03	1.03E+04	8.28E+03	**2.37E+03**
F23	Mean	2.92E+03(+)	2.85E+03(+)	3.07E+03(+)	3.04E+03(+)	3.17E+03(+)	2.81E+03(+)	2.88E+03(+)	3.14E+03(+)	**2.79E+03**
	Std	1.88E+01	4.86E+01	4.29E+01	8.33E+01	1.14E+02	**1.70E+01**	3.32E+01	1.37E+02	4.34E+01
	Med	2.92E+03	2.84E+03	3.07E+03	3.03E+03	3.15E+03	2.81E+03	2.89E+03	3.14E+03	**2.78E+03**
F24	Mean	3.06E+03(+)	2.99E+03(=)	3.26E+03(+)	3.21E+03(+)	3.29E+03(+)	2.97E+03(=)	3.07E+03(+)	3.39E+03(+)	**2.97E+03**
	Std	**1.33E+01**	6.16E+01	4.04E+01	9.91E+01	1.10E+02	1.47E+01	2.97E+01	2.17E+02	4.76E+01
	Med	3.07E+03	2.98E+03	3.26E+03	3.20E+03	3.26E+03	2.97E+03	3.06E+03	3.36E+03	**2.96E+03**
F25	Mean	3.90E+03(+)	2.97E+03(=)	3.54E+03(+)	3.36E+03(+)	3.38E+03(+)	3.02E+03(+)	2.96E+03(=)	3.17E+03(+)	**2.96E+03**
	Std	2.29E+02	3.19E+01	2.13E+02	2.49E+02	4.14E+02	**2.64E+01**	3.46E+01	9.78E+01	3.08E+01
	Med	3.87E+03	2.97E+03	3.51E+03	3.33E+03	3.20E+03	3.02E+03	2.96E+03	3.17E+03	**2.96E+03**
F26	Mean	6.78E+03(+)	5.10E+03(=)	7.82E+03(+)	6.70E+03(+)	8.65E+03(+)	5.37E+03(=)	6.03E+03(+)	7.14E+03(+)	**5.00E+03**
	Std	**1.48E+02**	1.34E+03	5.29E+02	1.19E+03	1.22E+03	3.39E+02	3.85E+02	1.59E+03	9.65E+02
	Med	6.76E+03	5.50E+03	7.80E+03	6.98E+03	8.70E+03	5.42E+03	6.09E+03	7.55E+03	**5.21E+03**
F27	Mean	3.35E+03(+)	3.35E+03(+)	3.57E+03(+)	3.53E+03(+)	3.49E+03(+)	3.25E+03(-)	**3.20E+03(-)**	3.27E+03(=)	3.27E+03
	Std	3.10E+01	6.80E+01	7.04E+01	8.41E+01	1.83E+02	7.50E+00	**8.77E-05**	1.09E+02	4.48E+01
	Med	3.34E+03	3.33E+03	3.57E+03	3.52E+03	3.40E+03	3.25E+03	**3.20E+03**	3.20E+03	3.26E+03
F28	Mean	3.91E+03(+)	3.34E+03(+)	4.54E+03(+)	4.04E+03(+)	3.97E+03(+)	3.41E+03(+)	**3.30E+03(-)**	3.72E+03(+)	3.32E+03
	Std	1.39E+02	3.41E+01	4.35E+02	4.86E+02	6.70E+02	5.60E+01	**4.68E-02**	2.66E+02	2.56E+01
	Med	3.91E+03	3.34E+03	4.42E+03	3.89E+03	3.76E+03	3.41E+03	**3.30E+03**	3.74E+03	3.32E+03
F29	Mean	4.82E+03(+)	4.45E+03(+)	5.21E+03(+)	4.61E+03(+)	4.88E+03(+)	**3.93E+03(-)**	4.21E+03(+)	5.04E+03(+)	4.05E+03
	Std	2.07E+02	2.90E+02	3.16E+02	3.47E+02	4.85E+02	**8.94E+01**	2.58E+02	5.81E+02	2.71E+02
	Med	4.82E+03	4.48E+03	5.19E+03	4.67E+03	4.90E+03	**3.95E+03**	4.24E+03	4.99E+03	4.05E+03
F30	Mean	6.80E+07(+)	9.60E+06(+)	2.08E+08(+)	3.03E+07(+)	3.42E+07(+)	5.34E+05	**9.43E+03(-)**	2.95E+07(+)	2.21E+05
	Std	3.31E+07	7.03E+06	6.20E+07	2.91E+07	1.19E+08	2.44E+05	**7.28E+03**	3.34E+07	1.47E+05
	Med	6.49E+07	8.93E+06	2.10E+08	2.31E+07	4.32E+06	4.96E+05	**6.12E+03**	1.56E+07	1.98E+05
	Total(+/=/-)	**30/0/0**	**22/8/0**	**30/0/0**	**29/1/0**	**28/2/0**	**23/4/3**	**19/4/7**	**29/1/0**	**-**

TABLE III

COMPARISON OF TEST RESULTS OF VARIOUS ALGORITHMS ON 30 DIMENSIONS OF CEC2013

Test function	Index	PSO	VPPSO	SCA	EDOLSCA	BKA	PLO	DE	BFGO	E-BFGO
F1	Mean	7.33E+03(+)	-1.39E+03(=)	1.91E+04(+)	8.10E+03(+)	1.48E+04(+)	-4.71E+01(+)	-9.12E+02(+)	3.69E+03(+)	**-1.40E+03**
	Std	1.31E+03	2.65E+01	3.57E+03	4.36E+03	1.02E+04	3.12E+02	5.11E+02	3.05E+03	**2.71E+00**
	Med	7.55E+03	**-1.40E+03**	1.89E+04	7.75E+03	1.28E+04	-7.13E+01	-1.02E+03	2.51E+03	-1.40E+03
F2	Mean	1.63E+08(+)	3.00E+07(+)	3.24E+08(+)	1.28E+08(+)	6.93E+07(+)	5.87E+07(+)	2.36E+07(=)	8.15E+07(+)	**2.03E+07**
	Std	5.10E+07	1.17E+07	1.15E+08	5.89E+07	1.01E+08	1.57E+07	1.06E+07	4.33E+07	**9.99E+06**
	Med	1.66E+08	2.67E+07	2.96E+08	1.21E+08	2.88E+07	5.50E+07	2.33E+07	7.05E+07	**1.79E+07**
F3	Mean	2.83E+10(+)	1.43E+10(+)	1.44E+11(+)	1.26E+12(+)	4.07E+13(+)	1.90E+10(+)	1.60E+10(+)	2.46E+11(+)	**7.53E+09**
	Std	8.35E+09	6.64E+09	1.16E+11	6.31E+12	2.18E+14	**3.84E+09**	8.89E+09	5.54E+11	4.04E+09
	Med	2.78E+10	1.31E+10	1.04E+11	6.64E+10	3.09E+10	1.88E+10	1.50E+10	5.94E+10	**6.13E+09**
F4	Mean	9.34E+04(+)	5.78E+04(+)	6.70E+04(+)	8.86E+04(+)	4.51E+04(+)	1.05E+05(+)	7.78E+04(+)	5.82E+04(+)	**3.31E+04**
	Std	2.21E+04	8.38E+03	1.29E+04	2.13E+04	1.15E+04	1.46E+04	1.74E+04	7.04E+03	**5.48E+03**
	Med	9.11E+04	6.03E+04	6.73E+04	8.92E+04	4.42E+04	1.10E+05	7.84E+04	5.88E+04	**3.45E+04**
F5	Mean	1.46E+03(+)	-7.78E+02(+)	3.33E+03(+)	5.83E+02(+)	3.14E+03(+)	-6.72E+02(+)	-4.08E+02(+)	9.15E+02(+)	**-8.51E+02**
	Std	5.08E+02	8.86E+01	9.88E+02	1.03E+03	4.00E+03	7.66E+01	7.52E+02	7.92E+02	**4.23E+01**
	Med	1.52E+03	-8.04E+02	3.53E+03	3.27E+02	2.03E+03	-6.74E+02	-6.30E+02	7.58E+02	**-8.55E+02**
F6	Mean	-2.97E+02(+)	-7.45E+02(+)	8.90E+02(+)	6.98E+01(+)	-2.05E+01(+)	-7.57E+02(+)	-7.78E+02(+)	-3.38E+02(+)	**-8.07E+02**
	Std	9.39E+01	7.45E+01	4.92E+02	5.23E+02	1.14E+03	**2.12E+01**	4.31E+01	2.37E+02	4.01E+01
	Med	-2.83E+02	-7.58E+02	9.10E+02	-4.77E+01	-4.04E+02	-7.53E+02	-7.79E+02	-4.10E+02	**-8.15E+02**
F7	Mean	-6.30E+02(+)	**-6.78E+02(=)**	-4.73E+02(+)	-4.87E+02(+)	3.07E+03(+)	-6.54E+02(+)	-6.32E+02(+)	1.97E+03(+)	-6.74E+02
	Std	3.16E+01	3.24E+01	1.90E+02	3.67E+02	1.86E+04	**1.57E+01**	4.68E+01	8.53E+03	4.20E+01
	Med	-6.33E+02	**-6.79E+02**	-5.27E+02	-5.87E+02	-5.71E+02	-6.56E+02	-6.44E+02	-4.86E+02	-6.79E+02
F8	Mean	-6.79E+02(=)	**-6.79E+02(=)**	-6.79E+02(=)	-6.79E+02(+)	-6.79E+02(=)	-6.79E+02(=)	-6.79E+02(=)	-6.79E+02(=)	-6.79E+02
	Std	6.10E-02	**3.99E-02**	4.88E-02	4.99E-02	5.17E-02	4.37E-02	7.68E-02	5.28E-02	5.72E-02
	Med	-6.79E+02	**-6.79E+02**	-6.79E+02	-6.79E+02	-6.79E+02	-6.79E+02	-6.79E+02	-6.79E+02	-6.79E+02
F9	Mean	-5.60E+02(+)	**-5.72E+02(-)**	-5.57E+02(+)	-5.63E+02(+)	-5.66E+02(+)	-5.68E+02(=)	-5.57E+02(+)	-5.62E+02(+)	-5.69E+02
	Std	1.76E+00	4.18E+00	**1.25E+00**	3.91E+00	3.79E+00	1.80E+00	1.55E+00	3.95E+00	2.48E+00
	Med	-5.60E+02	**-5.71E+02**	-5.57E+02	-5.63E+02	-5.65E+02	-5.68E+02	-5.57E+02	-5.61E+02	-5.70E+02
F10	Mean	7.81E+02(+)	-3.43E+02(+)	2.19E+03(+)	6.55E+02(+)	1.31E+03(+)	-1.27E+02(+)	-2.77E+02(+)	2.80E+02(+)	**-4.42E+02**
	Std	2.34E+02	8.35E+01	5.61E+02	4.23E+02	1.50E+03	8.61E+01	8.38E+01	3.20E+02	**2.72E+01**
	Med	8.13E+02	-3.48E+02	2.16E+03	6.51E+02	7.89E+02	-1.10E+02	-2.80E+02	2.41E+02	**-4.47E+02**
F11	Mean	-5.78E+01(+)	-1.46E+02(+)	6.35E+01(+)	-3.18E+01(+)	1.76E+02(+)	-2.25E+02(+)	-2.21E+02(+)	5.69E+01(+)	**-2.62E+02**
	Std	2.52E+01	8.84E+01	4.74E+01	8.03E+01	1.20E+02	**2.01E+01**	3.66E+01	1.42E+02	3.59E+01
	Med	-5.98E+01	-1.66E+02	6.52E+01	-4.19E+01	1.80E+02	-2.24E+02	-2.24E+02	3.03E+01	**-2.69E+02**
F12	Mean	4.24E+01(+)	-1.69E+01(+)	1.76E+02(+)	1.23E+02(+)	3.23E+02(+)	-6.08E+01(+)	-5.01E+01(+)	1.06E+02(+)	**-1.21E+02**
	Std	2.68E+01	7.95E+01	4.24E+01	6.69E+01	1.12E+02	**2.39E+01**	2.61E+01	1.16E+02	5.95E+01
	Med	4.19E+01	-1.79E+01	1.72E+02	1.12E+02	3.30E+02	-5.55E+01	-4.47E+01	8.02E+01	**-1.29E+02**
F13	Mean	1.49E+02(+)	1.81E+02(+)	2.89E+02(+)	2.36E+02(+)	4.24E+02(+)	8.85E+01(+)	5.55E+01(=)	2.61E+02(+)	**5.03E+01**
	Std	**1.95E+01**	7.42E+01	3.96E+01	7.90E+01	1.05E+02	2.46E+01	2.38E+01	9.84E+01	5.65E+01
	Med	1.51E+02	1.68E+02	2.93E+02	2.17E+02	4.49E+02	8.32E+01	5.24E+01	2.34E+02	**4.70E+01**
F14	Mean	7.84E+03(+)	4.19E+03(+)	7.71E+03(+)	6.77E+03(+)	5.09E+03(+)	4.10E+03(+)	7.92E+03(+)	5.24E+03(+)	**1.88E+03**
	Std	3.23E+02	7.67E+02	3.13E+02	6.66E+02	9.16E+02	**2.48E+02**	3.96E+02	7.16E+02	4.59E+02
	Med	7.89E+03	4.14E+03	7.74E+03	6.75E+03	5.01E+03	4.07E+03	7.93E+03	5.26E+03	**1.89E+03**
F15	Mean	8.07E+03(+)	4.40E+03(=)	8.22E+03(+)	7.61E+03(+)	4.80E+03(=)	6.31E+03(+)	8.54E+03(+)	6.78E+03(+)	**4.31E+03**
	Std	3.15E+02	8.98E+02	2.94E+02	4.01E+02	8.87E+02	**2.62E+02**	3.22E+02	8.45E+02	9.32E+02
	Med	8.07E+03	4.27E+03	8.26E+03	7.65E+03	4.84E+03	6.21E+03	8.61E+03	6.94E+03	**4.26E+03**
F16	Mean	2.03E+02(+)	**2.01E+02(-)**	2.03E+02(+)	2.04E+02(+)	2.03E+02(+)	2.02E+02(=)	2.04E+02(+)	2.03E+02(+)	2.02E+02
	Std	4.19E-01	3.85E-01	4.02E-01	4.38E-01	5.00E-01	**2.85E-01**	3.91E-01	6.20E-01	6.71E-01
	Med	2.04E+02	**2.01E+02**	2.03E+02	2.04E+02	2.03E+02	2.02E+02	2.04E+02	2.03E+02	2.02E+02
F17	Mean	1.08E+03(+)	6.57E+02(+)	9.44E+02(+)	8.38E+02(+)	1.00E+03(+)	6.62E+02(+)	5.76E+02(+)	8.33E+02(+)	**5.04E+02**
	Std	1.18E+02	1.12E+02	6.64E+01	9.53E+01	9.35E+01	4.05E+01	**2.68E+01**	1.20E+02	4.94E+01
	Med	1.09E+03	6.41E+02	9.29E+02	8.57E+02	1.01E+03	6.62E+02	5.72E+02	8.43E+02	**4.96E+02**
F18	Mean	1.17E+03(+)	7.49E+02(+)	1.04E+03(+)	9.51E+02(+)	1.10E+03(+)	8.28E+02(+)	6.94E+02(+)	8.68E+02(+)	**6.42E+02**
	Std	1.03E+02	9.85E+01	6.83E+01	1.07E+02	1.22E+02	3.73E+01	**2.53E+01**	8.93E+01	5.13E+01
	Med	1.21E+03	7.38E+02	1.03E+03	9.58E+02	1.13E+03	8.36E+02	6.85E+02	8.65E+02	**6.48E+02**
F19	Mean	1.51E+03(+)	5.36E+02(+)	4.20E+04(+)	1.23E+04(+)	3.84E+04(+)	5.66E+02(+)	1.42E+03(+)	5.96E+03(+)	**5.22E+02**
	Std	5.30E+02	1.95E+01	5.38E+04	2.58E+04	9.76E+04	3.13E+01	1.49E+03	7.37E+03	**8.92E+00**
	Med	1.38E+03	5.33E+02	2.07E+04	3.32E+03	1.09E+04	5.54E+02	7.59E+02	2.82E+03	**5.20E+02**
F20	Mean	6.15E+02(+)	6.15E+02(+)	6.15E+02(+)	6.15E+02(+)	6.15E+02(+)	6.15E+02(+)	6.14E+02(+)	6.15E+02(+)	**6.14E+02**
	Std	4.67E-01	3.07E-01	2.80E-01	**2.65E-01**	2.81E-01	4.71E-01	5.37E-01	3.08E-01	7.80E-01
	Med	6.15E+02	6.15E+02	6.15E+02	6.15E+02	6.15E+02	6.15E+02	6.14E+02	6.15E+02	**6.14E+02**
F21	Mean	2.53E+03(+)	1.37E+03(+)	2.90E+03(+)	2.36E+03(+)	2.67E+03(+)	1.69E+03(+)	1.49E+03(+)	2.32E+03(+)	**1.10E+03**
	Std	3.00E+02	3.01E+02	9.67E+01	3.07E+02	2.44E+02	1.44E+02	3.88E+02	3.17E+02	**6.10E+01**
	Med	2.65E+03	1.30E+03	2.92E+03	2.37E+03	2.71E+03	1.68E+03	1.38E+03	2.39E+03	**1.14E+03**
F22	Mean	8.82E+03(+)	5.91E+03(+)	9.26E+03(+)	8.14E+03(+)	6.91E+03(+)	5.55E+03(+)	9.07E+03(+)	6.85E+03(+)	**3.34E+03**
	Std	3.97E+02	9.81E+02	5.00E+02	8.54E+02	1.09E+03	4.18E+02	**3.50E+02**	1.20E+03	6.62E+02
	Med	8.81E+03	5.75E+03	9.34E+03	8.22E+03	7.14E+03	5.61E+03	9.14E+03	6.75E+03	**3.24E+03**
F23	Mean	9.17E+03(+)	**5.97E+03(=)**	9.53E+03(+)	8.72E+03(+)	6.75E+03(+)	7.59E+03(+)	9.47E+03(+)	7.97E+03(+)	6.27E+03
	Std	3.82E+02	7.02E+02	**3.54E+02**	6.65E+02	9.68E+02	5.08E+02	4.57E+02	9.66E+02	9.58E+02
	Med	9.18E+03	**5.89E+03**	9.66E+03	8.79E+03	6.70E+03	7.69E+03	9.54E+03	8.01E+03	6.20E+03
F24	Mean	1.31E+03(+)	**1.29E+03(=)**	1.33E+03(+)	1.32E+03(+)	1.32E+03(+)	1.29E+03(=)	1.31E+03(+)	1.30E+03(+)	1.29E+03
	Std	5.02E+00	1.01E+01	5.79E+00	1.33E+01	2.54E+01	4.13E+00	**3.71E+00**	1.04E+01	1.13E+01
	Med	1.31E+03	1.29E+03	1.33E+03	1.32E+03	1.32E+03	**1.29E+03**	1.31E+03	1.30E+03	1.29E+03
F25	Mean	1.42E+03(+)	1.40E+03(=)	1.44E+03(+)	1.45E+03(+)	1.43E+03(+)	**1.40E+03(-)**	1.41E+03(=)	1.40E+03(=)	1.40E+03
	Std	5.40E+00	9.40E+00	6.32E+00	1.41E+01	1.40E+01	5.27E+00	**2.89E+00**	9.30E+00	1.33E+01
	Med	1.42E+03	1.40E+03	1.44E+03	1.45E+03	1.43E+03	**1.40E+03**	1.41E+03	1.40E+03	1.40E+03
F26	Mean	1.53E+03(+)	1.52E+03(+)	1.45E+03(-)	1.46E+03(-)	1.56E+03(+)	**1.41E+03(=)**	1.56E+03(+)	1.55E+03(+)	1.47E+03
	Std	8.93E+01	8.05E+01	5.68E+01	7.72E+01	7.42E+01	**2.28E+00**	8.22E+01	8.85E+01	8.54E+01
	Med	1.60E+03	1.57E+03	1.43E+03	1.41E+03	1.59E+03	1.41E+03	1.61E+03	1.60E+03	**1.40E+03**
F27	Mean	2.61E+03(+)	**2.39E+03(-)**	2.74E+03(+)	2.58E+03(+)	2.58E+03(+)	2.43E+03(=)	2.68E+03(+)	2.65E+03(+)	2.46E+03
	Std	5.66E+01	8.75E+01	4.23E+01	1.15E+02	9.90E+01	3.99E+01	**3.34E+01**	9.36E+01	9.14E+01
	Med	2.62E+03	**2.40E+03**	2.75E+03	2.57E+03	2.59E+03	2.45E+03	2.68E+03	2.66E+03	2.46E+03
F28	Mean	3.95E+03(+)	3.95E+03(+)	4.86E+03(+)	4.37E+03(+)	6.16E+03(+)	3.13E+03(+)	2.71E+03(+)	5.47E+03(+)	**2.48E+03**
	Std	2.97E+02	1.05E+03	3.45E+02	4.74E+02	5.39E+02	**1.31E+02**	3.80E+02	9.14E+02	1.01E+03
	Med	3.88E+03	4.27E+03	4.82E+03	4.28E+03	6.15E+03	3.15E+03	2.70E+03	5.49E+03	**1.92E+03**
	Total(+/=/-)	**27/1/0**	**18/7/3**	**26/1/1**	**27/0/1**	**26/2/0**	**21/6/1**	**24/4/0**	**26/2/0**	**-**

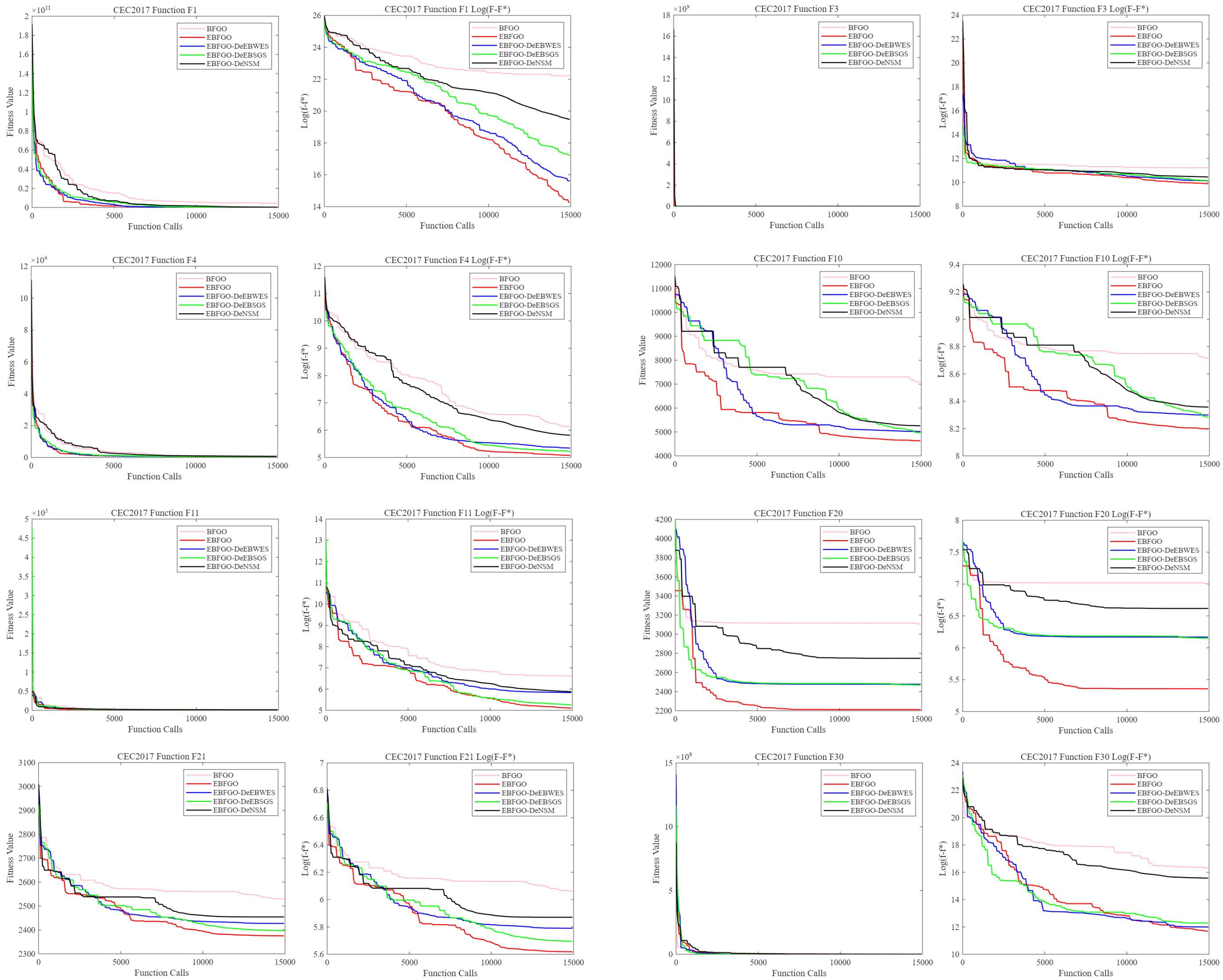

Fig. 3. Convergence processes of E-BFGO algorithm variants and BFGO algorithm on 30 dimensions of CEC2017

According to the experimental results from both test sets, the E-BFGO algorithm achieves the best performance in terms of mean and median values. The Wilcoxon rank-sum test further demonstrates that E-BFGO exhibits significant superiority over other compared algorithms, indicating its excellent performance in solution accuracy. However, E-BFGO does not show outstanding results on the standard deviation metric, particularly on composite function tests, suggesting insufficient robustness in complex scenarios. This may be attributed to defects introduced by certain stochastic strategies, yet it is also a necessary trade-off for the algorithm to escape local stagnation.

As illustrated in the Fig. 3, the synergistic effect of the three extension strategies yields optimal performance. EBFGO-DeNSM tends to cause algorithmic stagnation and a severe decline in search performance when confronted with complex landscapes characterized by multimodal, hybrid, and composite functions that are riddled with local traps. This confirms that the natural selection mechanism is a critical strategy within the extended framework, enabling the algorithm to escape local optima and attain high-quality solutions. Both EBFGO-DeEBWES and EBFGO-DeEBSGS exhibit slightly insufficient search precision, lacking more refined exploration. This also indicates that the extended extension strategy and the extended growth strategy effectively enhance the traversal efficiency of the algorithm across vast solution spaces while contributing to local exploitation capabilities. Overall, the effectiveness of all three extension strategies is validated.

V. Conclusion

Inspired by the growth characteristics of bamboo forests and natural laws, this paper comprehensively extends and modifies the original BFGO algorithm while preserving some of its mechanisms, and introduces necessary new strategies. Through experimental comparisons, the improved algorithm's strategies have each played their intended roles, enhancing the performance of the original algorithm. On both the CEC2017 and CEC2013 test sets, the E-BFGO algorithm significantly outperforms other compared algorithms in terms of solution accuracy. Although its stability is not the best, in terms of overall capability, E-BFGO remains highly competitive in terms of overall performance.

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